Giovanni Sambo

Aeronautical Engineer

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Summary

MSc Aeronautical Engineer with a broad applied background: wind tunnel testing, CFD, wind energy and industrial flight testing. Thesis on the Flettner rotor for wind-assisted ship propulsion (WASP), combining a wind tunnel campaign with OpenFOAM CFD validation on an HPC cluster. Interested in renewable energy and maritime decarbonisation. Available immediately, open to relocation in Europe.

Education

2023 - 2026 **MSc Aeronautical Engineering**, *Politecnico di Milano*, Milan, Italy, Graduated October 2026
Aerodynamics, CFD, experimental fluid dynamics, wind & energy systems, flight testing, propulsion.

Jan - Jun 2024 **Erasmus Exchange**, *ETSi, Univ. de Sevilla*, Seville, Spain
Heat Transfer, Aircraft Trajectory Optimization.

2020 - 2023 **BSc Aerospace Engineering**, *Politecnico di Milano*, Milan, Italy

Master's Thesis

Title *Wind Tunnel and Numerical Study of a Finite Flettner Rotor at High Reynolds Numbers for Wind-Assisted Ship Propulsion*

Experimental **Wind tunnel campaign** (GVPM, PoliMi) on a scaled Flettner rotor across critical and supercritical Reynolds regimes, varying spin ratio and endplate geometry. Strain-gauge force balance, rotating surface-pressure measurements, high-speed smoke visualization, uncertainty analysis in MATLAB.

Numerical 2D and 3D RANS/URANS in **OpenFOAM** on the PoliMi **HPC cluster**: SST vs SST-LM transition model, mesh convergence, wind tunnel confinement effects, validation against experiments and literature.

Academic Projects

2025 - 2026 **Flight Testing: Pipistrel Virus SW121A**, *Politecnico di Milano & Pipistrel Italia*
Industrial campaign: flight card design and mission planning, DEWESoft DAS, in-flight data acquisition, post-flight analysis of stall and performance.

2025 - 2026 **2.3 MW IEC Class IA Wind Turbine Design**, *Politecnico di Milano*
Aeroelastic design in Cp-Lambda (BEM). Structural sizing via BECAS/RotorCS, TurbSim wind fields (IEC 61400-1), PID gain-scheduling, DLC load assessment. CoE minimization over 8 design variants.

2024 - 2025 **Experimental Fluid Dynamics: Wind Tunnel Labs**, *Politecnico di Milano*
Three **wind tunnel** campaigns (PoliMi): force balance on NACA 0015; pressure and wake survey on NACA 23012; PIV of side-by-side eVTOL rotor configurations.

2024 - 2025 **Aerodynamic Analysis of a Travel Bus**, *Politecnico di Milano*
Drag reduction via OpenFOAM, validated against **wind tunnel** measurements (PoliMi). Combined CFD and experimental approach.

2024 - 2025 **CFD Analysis: Formula 1 Front Wing**, *Politecnico di Milano*
2D compressible RANS/URANS in SU2. Ground effect and porpoising (6 Hz). SA vs SST k-ω. Post-processing in ParaView.

2024 - 2025 **Crosswind Rollover: Heavy Truck**, *Politecnico di Milano*
Multi-DOF MATLAB model with Pacejka tyres under transient aerodynamic crosswind loading.

Work Experience

Jun - Jul 2019 **Assistant Aircraft Designer, Internship**, *Superjet Sukhoi*
Maintenance documentation and data archiving for the Superjet Sukhoi 100.

2018 - 2022 **Lifeguard, salesperson, waiter**, *Turismo Srl · Air Food Srl · Gattopardo*

Skills & Languages

CFD & Aero OpenFOAM, SU2, XFOIL, ParaView
Experimental Wind tunnel, force balance, PIV
Programming MATLAB (adv.), Python (scripting), C
Languages Italian (native), English (professional), Spanish (basic)
Wind Turbine Cp-Lambda, BECAS, RotorCS, TurbSim
CAD & DAQ SolidWorks, Inventor, DEWESoft
HPC Linux cluster (SGE), Bash
Sport: competitive water polo, team captain